of these interrelationships, and because his equations showed that electric and magnetic fields are as entwined as the fibers in a Rastafarian's dreadlocks, they were eventually christened *electromagnetic* fields, and the influence they exert the *electromagnetic* force.

Today, we are constantly immersed in a sea of electromagnetic fields. Your cellular telephone and car radio work over enormous expanses because the electromagnetic fields broadcast by telephone companies and radio stations suffuse impressively wide regions of space. The same goes for wireless Internet connections; computers can pluck the entire World Wide Web from electromagnetic fields that are vibrating all around us—in fact, right through us. Of course, in Maxwell's day, electromagnetic technology was less developed, but among scientists his feat was no less recognized: through the language of fields, Maxwell had shown that electricity and magnetism, although initially viewed as distinct, are really just different aspects of a single physical entity.

Later on, we'll encounter other kinds of fields—gravitational fields, nuclear fields, Higgs fields, and so on—and it will become increasingly clear that the field concept is central to our modern formulation of physical law. But for now the critical next step in our story is also due to Maxwell. Upon further analyzing his equations, he found that changes or disturbances to electromagnetic fields travel in a wavelike manner at a particular speed: 670 million miles per hour. As this is precisely the value other experiments had found for the speed of light, Maxwell realized that light must be nothing other than an electromagnetic wave, one that has the right properties to interact with chemicals in our retinas and give us the sensation of sight. This achievement made Maxwell's already towering discoveries all the more remarkable: he had linked the force produced by magnets, the influence exerted by electrical charges, and the light we use to see the universe—but it also raised a deep question.

When we say that the speed of light is 670 million miles per hour, experience, and our discussion so far, teach us this is a meaningless statement if we don't specify relative to *what* this speed is being measured. The funny thing was that Maxwell's equations just gave this number, 670 million miles per hour, without specifying or apparently